

Worksheet W11: Optimization

"Using derivatives to find the 'best' or 'most.'"

Chapter 11 · 12 Problems · Name: _____ Date: _____

To find the maximum or minimum: Find $f'(x)$, set it to zero, and solve.

Problem 1

Question: What is **optimization** in calculus?

💡 *Hint: It's about finding the best price, shortest time, or greatest profit.*

Problem 2

Question: Why is the **derivative zero** at both maximums and minimums?

💡 *Hint: Think about the top of a hill or the bottom of a valley.*

Problem 3

Find the critical point of $f(x) = x^2 - 4x$.

💡 *Hint: Find $f'(x)$, set it to zero, and solve.*

Problem 4

For $f(x) = -x^2 + 4x$, find the maximum value.

💡 Hint: Find the critical point, then plug it into $f(x)$.

Problem 5

For $f(x) = x^2 - 6x + 10$, find the minimum value.

💡 Hint: Find the critical point, then plug it into $f(x)$.

Problem 6

Question: What is the **First Derivative Test**?

💡 Hint: It uses the sign of the derivative before and after the critical point.

Problem 7

Question: What is the **Second Derivative Test**?

💡 Hint: It uses the second derivative at the critical point.

Problem 8

A company's profit is $P(x) = -x^2 + 10x - 16$, where x is the number of units sold. Find the number of units that maximizes profit.

💡 Hint: Find $P'(x)$, set to zero, and solve.

Problem 9

A farmer has 100 meters of fencing to enclose a rectangular field. What dimensions maximize the area?

💡 Hint: $2x + 2y = 100 \rightarrow y = 50 - x$. Area = $x(50 - x)$.

Problem 10

A box with a square base and an open top must have a volume of 32 cubic feet. What dimensions minimize the amount of material used?

💡 Hint: $SA = x^2 + 4xh$, where $h = 32/x^2$.

Problem 11

A company sells a product. The demand is given by Price = $100 - 2x$, where x is the number of units sold (in thousands). Find the number of units that maximizes revenue.

💡 Hint: $R(x) = x(100 - 2x) = 100x - 2x^2$.

Problem 12

Question: What does the **Mental Zoom** show us about optimization?

💡 Hint: Zoom in on the top of a hill — what does the curve look like?